

Unveiling the Inherent Needs: GPT Builder as Participatory Design Tool for Exploring Needs and Expectation of AI with Middle-Aged Users

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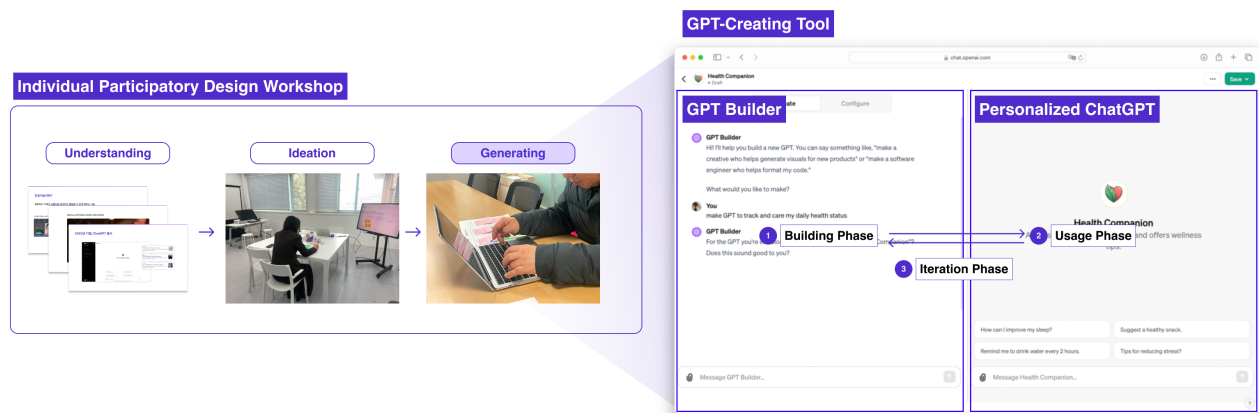


Figure 1: Overview of individual participatory design workshop and the process of utilizing GPT-creating tool

ABSTRACT

A generative session that directly involves users in the design process is an effective way to design user-centered experiences by uncovering intrinsic needs. However, engaging users who lack coding knowledge in AI system design poses significant challenges. Recognizing this, the recently revealed GPT-creating tool, which allows users to customize ChatGPT through simple dialog interactions, is a promising solution. We aimed to identify the possibility of using this tool to uncover intrinsic users' needs and expectations towards AI. We conducted individual participatory design workshops with generative sessions focusing on middle-aged individuals. This approach helped us to delve into the latent needs and expectations

of conversational AI among this demographic. We discovered a wide range of unexpressed needs and expectations for AI among them. Our research highlights the potential and value of using the GPT-creating tool as a design method, particularly for revealing the users' unexpressed needs and expectations.

CCS CONCEPTS

• Human-centered computing → User studies.

KEYWORDS

GPT, Participatory Design (PD), Generative Design, Inherent needs, Artificial Intelligence (AI), LLM-based CAS

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1 INTRODUCTION AND BACKGROUND

As Artificial Intelligence (AI) becomes more prevalent in our lives, designing user experiences of AI that align with the behaviors and needs of users has become an important task for designers. However, designers often struggle understanding users' behaviors and needs because AI systems can behave unexpectedly, learn and evolve, and generate complex outputs. These challenges lead to the difficulty of designing user-centered experiences with AI [2, 7, 16].

To design user-centered experiences by uncovering the intrinsic needs of users, Sanders and Stappers have suggested a generative session that involves users directly in the design process to encourage them to reveal their desires [11]. In this regard, such a generative session approach could be particularly valuable for exploring user behavior and needs in the context of AI system experience design. However, engaging users in the design process of AI systems, especially through generative sessions, presents a significant challenge. This is because involving users without coding skills in the AI design process makes it even more difficult for users to articulate their needs and behaviors.

On November 6, 2023, Open AI released a GPT-creating tool, allowing users to customize ChatGPT for specific purposes [9]. This feature allows even those without coding knowledge to create their own ChatGPT through simple dialog interaction with the GPT builder. We assumed that creating users' own ChatGPT utilizing GPT-creating tool would be an approach as a generative session to support designers in understanding users' intrinsic needs and expectations.

With this background, we aimed to explore opportunities for utilizing a GPT-creating tool to uncover the intrinsic user needs and expectations through the generative session. To achieve this, among the various user groups, we focused on middle-aged individuals who have limited experience using ChatGPT as a case study. Middle-aged adults have the basic ability to use digital devices, enabling them to have conversations with a GPT Builder. However, as they are not digital natives, some of them find it challenging to learn and use new technologies like AI [5] and struggle to express how they desire to utilize them clearly.

Moreover, middle-aged adults are in a period of facing significant personal and social changes [15], and it is crucial to recognize their challenges and provide assistance, not only for personal but also for substantial social significance. Given that Large language model (LLM)-based Conversational Agents (CAs), represented by ChatGPT, serve as effective tools for providing extensive information on daily issues [12], as well as offering personalized responses [17], this characteristic underscores their potential in closely supporting middle-aged individuals. While numerous studies have identified the potential of LLM-based CAs in supporting younger [1] or older generations [10], few have focused on the inclusion of the middle-aged demographic in this research field, leading us to expect new and intriguing insights by selecting them as the subjects of our research.

In this research, we conducted an individual participatory design workshop including generative sessions with eleven middle-aged individuals, using a GPT-creating tool to craft their personalized ChatGPT. We uncovered their intrinsic needs and expectations,

and then further discussed the potential of a GPT-creating tool for generative sessions.

2 METHOD

Our research aimed to use the GPT-creating tool to uncover the intrinsic needs and expectations of middle-aged adults regarding conversational AI and identify the opportunities of this tool. To accomplish this, we conducted a participatory design approach, including a generative session in which participants crafted personalized ChatGPT. We believed that the participants' intrinsic needs and expectations would especially be discovered while observing the dialogue with the GPT builder and their usage of personalized ChatGPT. Therefore, to closely examine the generating process, the experiment was conducted in a one-on-one, face-to-face setting, with each participant spending approximately two hours on the task.

2.1 Participants

Participants were recruited through a local online platform in South Korea. Since the participants were required to create agents using GPT-creating tool, we selected individuals who 1) had no problem reading and writing and 2) were familiar with computer typing. To explore a broad range of issues of the middle ages and prevent bias regarding occupational status, we verified their job status during recruitment to ensure diversity among the participants. Ultimately, we recruited 12 participants, aged between 48 and 63 years, and due to the absence of one participant, 11 participants (five male and six female) attended the individual participatory design session (Table 1). All participants received informed consent before beginning the study.

2.2 Study Procedure

The participatory design consisted of three sessions: understanding, ideation, and generating. In the understanding session, we concentrated on sparking middle-aged participants' interest and uniformly aligning their comprehension regarding AI. To accomplish this, participants watched a four-minute scene from the movie *'Her'* [3], where the main character interacts with the AI, Samantha. This was intended to spark interest and facilitate ice-breaking through observing human-AI interaction. Then, the brief explanation covered AI technologies that can be encountered in daily life. Considering that 8 out of the 11 participants had never used ChatGPT before and none of them used ChatGPT frequently, we arranged a 10-minute session for them to use ChatGPT. This aimed to assist participants with limited experience in using ChatGPT to grasp how it interacts with users.

Participants were provided with worksheets in the ideation session to brainstorm 'concerns and interests of the middle ages' and 'possible AI solutions that could be addressed'. The goal of this session was to explore a broad spectrum of concerns and interests and to brainstorm various potential AI services. Participants spent 10 minutes brainstorming their concerns and interests, followed by 15 minutes dedicated to ideating as many service ideas as possible.

In the final generating session, ideas from the earlier sessions were shared, and one or two ideas were selected to create their own personalized ChatGPT. Since the experiment was conducted

Table 1: Informations of the participants

ID	Age	Gender	Occupational Status
P1	63	Male	Employed in part-time or temporary position
P2	61	Female	A full-time housewife, not employed outside
P3	53	Female	A full-time housewife, not employed outside
P4	56	Male	Employed in part-time or temporary position
P5	60	Male	Currently employed full-time with no plans for retirement
P6	58	Male	Currently employed full-time with no plans for retirement
P7	54	Female	Self-employed or working as a freelancer
P8	48	Female	Self-employed or working as a freelancer
P9	48	Female	A full-time housewife, not employed outside
P10	51	Male	Currently employed full-time with no plans for retirement
P11	53	Female	Currently employed full-time with plans for retirement

with Korean participants, the GPT builder was pre-arranged in Korean, the language in which the participants felt most comfortable communicating. Participants developed a personalized ChatGPT through conversation with the GPT builder. During and after the experiment, the researcher closely observed the design process and conducted follow-up interviews to delve deeper into it.

2.3 Data Analysis

The first author transcribed worksheets that participants generated during an ideation session, the dialogue records from the process of creating personalized ChatGPT with the GPT builder, and recorded follow-up interviews from individual participatory design workshops. Participants formulated 62 ideas, and 19 were designed into personalized ChatGPT. The researchers comprehensively verified all the research data to ensure that the nuances of the data were accurately understood without misinterpretation. Subsequently, we used the open and axial coding method as a thematic coding approach [4], which followed the practice of similar qualitative studies in HCI. The initial coding was completed using an inductive approach, focusing on the needs and expectations related to AI. The researchers used axial coding to group codes into categories and organized the initially extracted codes to derive themes according to middle-aged adults' challenges. We believed that these themes would best convey the findings. After that, the researcher analyzed the dialogues that participants had with the GPT-creating tool and transcripts of recorded interviews again to identify the potential of the GPT-creating tool to identify users' intrinsic needs through the iterative session. Using the open and axial coding as before, researchers grouped codes into categories and extracted themes related to the process of GPT-creating usage.

3 FINDINGS AND DISCUSSIONS

The primary objective of this study was to investigate the opportunities of using the GPT-creating tool as a means to reveal users' intrinsic needs and expectations within a generative session, focusing on middle-aged adults as a case study. This section will mainly report the findings from our exploration of the needs and

expectations of using conversational AI in the lives of middle-aged adults. Subsequently, we would like to delve into the strengths and potential of a GPT-creating tool to uncover the inherent needs of the users.

3.1 The Intrinsic Expectations for conversational AI of the Middle-Aged Adults

During the individual participatory design process, middle-aged participants expressed a variety of issues and concerns encountered in their lives and showed interest in receiving support through conversational AI. In the process of designing a personalized ChatGPT, they more concretely revealed what roles they expect from conversational AI and the specific reasons for these expectations. In alignment with previous papers [13–15], discussing various tasks faced by the middle-aged in their developmental process, our study identified a total of four areas of challenges they confronted: adjusting to the body's physical changes; being sandwiched between young and old generations; reorganizing living arrangements; and ensuring economic security for old age. These challenges led to the need to resolve them, which further translated into expectations for AI services, detailed in Table 2.

3.1.1 Managing the Body's Physical Changes. As individuals enter the middle ages, they experience physical and physiological changes, unlike in their younger age [15]. As age-related ailments such as hypertension, diabetes, and prostate enlargement developed, they began to feel the need to manage these chronic diseases. This led to expectations of AI capable of precisely identifying and managing my chronic conditions (P1). Even without suffering from particular diseases, increased attention to health arises as they observe their aging bodies and close ones falling ill. Participants expected AI to manage their daily health, monitor their everyday condition (P2), recommend healthy diets (P2, P9), and even predict and diagnose based on their health status (P3, P10). They commonly expected AI to serve a role akin to a personalized physician who manages their health data. These physical changes also raised concerns about handling emergencies in old age. Participants desired

Table 2: Explored inherent needs related to challenges faced by middle-aged individuals and their expected support from AI

Challenges of Middle-Aged Individuals and Expected AI Support	
1. Managing the Body's Physical Changes (see chapter 3.1.1)	
<i>Needs</i>	<i>How can AI support you?</i>
1.1 Fear of the new diseases found while getting old emerged, leading to a need for disease management.	- Identify emergencies when alone at home and make emergency contacts (P8, P9)
1.2 Observing their aging bodies and the illness of their close ones, awareness of daily health management increases.	- Monitor my health status every morning (P2) - Recommend guidance on healthy diets (P2, P9) - Track my daily health status, predict and diagnose diseases (P3, P10)
1.3 Concerns about not being able to handle emergency situations (such as solitary death) arise with age	- Identify emergencies when alone at home and make emergency contacts (P8, P9)
2. Sharing Challenges Arising from Being the Sandwiched Generation (see chapter 3.1.2)	
<i>Needs</i>	<i>How can AI support you?</i>
2.1 Seek to get advice on concerns arises from interpersonal relationships	- Provide counseling on interpersonal relationships; especially with children, parents and spouses (P6, 10)
2.2 Want to have a conversation with someone other than their spouse or family.	- Listen to and discuss my thoughts from my perspective (P7)
2.3 Struggling to fulfill my ideal role in interpersonal relationships.	- Take over the roles that I am supposed to do (P11)
3. Guiding to Reorganize Living Arrangements (see chapter 3.1.3)	
<i>Needs</i>	<i>How can AI support you?</i>
3.1 Desiring to find a second career after retirement	- Suggest economic activities suitable for my situation (P6, P10)
3.2 Contemplating how to spend the suddenly increased amount of leisure time.	- Recommend various activities for leisure time (P4, P5) - Instruct leisure activities about what they are interested in (P3, P4, P6)
4. Preparing Economic Security for Old Age (see chapter 3.1.4)	
<i>Needs</i>	<i>How can AI support you?</i>
4.1 Concern about the adequacy of financial resources for old age	- Plan how to utilize current assets in the future (P9) - advice on investment to increase current assets (P1, P6, P10)

AI to monitor their condition alone at home and make emergency calls if needed (P8, P9).

3.1.2 Sharing Challenges Arising from Being the Sandwiched Generation. Middle-aged individuals often find themselves as the 'sandwiched generation' between the young and old generations within the family. They typically face the dual responsibility of supporting their children's independence while caring for aging parents [15]. They also coincide with redefining spousal relationships [13]. These interrelational changes have led to a desire for advice and counseling, expecting AI to provide professional guidance on relationships with children, parents, and spouses (P6, P10). Alongside this, there was a desire to share deeper inner thoughts without professional counseling. P7, for example, created a chatGPT to chat that she couldn't discuss with their spouse or friends. Interestingly, since the middle ages are aware of their ideal role based on life experiences,

the stress of not fulfilling these roles adequately leads to an expectation that AI can take over these roles. P11, dealing with a mother showing symptoms of dementia, built a chatGPT to converse with her mother on her behalf, following the basic rules for conversing with patients exhibiting symptoms of dementia. They expected AI to perform task-oriented roles [8], such as providing information and analysis data, as well as social roles, replacing human roles and communicating.

3.1.3 Guiding to Reorganize Living Arrangements. As middle-aged individuals face or approach retirement, they are often required to reorganize their life patterns. The increased life expectancy also allowed them to grow their desire to explore second careers. The desire for a new career led to expectations that AI could analyze personal information, such as individual financial status or interests, to recommend suitable qualifications or jobs (P6, P10). Moreover, they faced challenges in how to spend increased leisure time, which was

different from their active working years. Some expressed a desire to learn things they had not before, asking AI for recommendations on what to learn and how to spend leisure time (P4, P5) or instructing them about what they are interested in, such as cooking or rural living (P3, P4, P6). We also found an expectation for AI to assist in making leisure activities more comfortable, such as planning trips tailored to one's situation (P10), rescheduling activities based on health status during travels (P3), or supporting hobbies like reading books and golf (P4).

3.1.4 Preparing Economic Security for Old Age. The middle ages are approaching a stage of life that involves occupational and situational changes and preparing for old age, so they are concerned about securing financial stability after retirement. This concern has led to a demand for AI to provide detailed plans on utilizing their current assets (P9) and even advice on specific investment strategies to increase their wealth (P1, P6, P10). They expected AI to serve the role of a personalized financial advisor or counselor optimized for the individual.

3.2 Opportunities of User-Led AI Creation as a Method to Understand Latent User Needs

During our study, we discovered various tacit needs, conscious but unexpressed desires and latent needs, which are subconscious needs that cannot be articulated [11], while using the GPT-creating tool in generative session. Instead of merely asking users to verbally express their expectations, the process of creating their desired conversational AI using the GPT-creating tool naturally uncovered their inherent needs and expectations that were dormant in their consciousness. The process of creating a personalized ChatGPT with the GPT-creating tool consisted of three steps: 1) the *Building phase*, where participants engaged in dialogue with the GPT builder, 2) the *Usage phase*, where they experimented with the personalized ChatGPT, and 3) the *Iteration phase*, where they revisited the GPT builder for modifications with further dialogue. In this session, we will delve into the opportunities and potential of the GPT-creating tool to uncover inherent needs and expectations discovered through the generating process.

3.2.1 Unexpressed Thoughts Revealed in the Dialogue with GPT Builder. GPT Builder inquires, "What would you like to make?" during the building phase. As users respond to this question, they can refine their unarticulated expectations. For example, P1 initially expressed a desire for "AI that proposes a stable financial management plan" in the first ideation phase, but through dialogue with GPT Builder, refined this idea into "AI that suggests a plan for managing life expenses with the national pension received from 2023, while saving and investing a portion to counter inflation and health issues for stable and efficient returns." Furthermore, GPT Builder asks users about their preferences for the tone of conversation or the type of conversational style. This process enables users to easily define the persona of their agents, which has not yet been defined. For instance, P11, while creating a ChatGPT to converse with her mother, requested features like "friendly and simple conversation, not too lengthy, using informal language just like a real friend." The characteristics of natural language-based interaction are known to

involve ambiguity, and sometimes multiple interactions are necessary to clarify the true intent [6]. Since the GPT builder is also a model based on the LLM system, participants had to clarify their intentions through dialogue. Thus, they were able to express their hidden needs accurately. Users often hold their desires within their consciousness, yet they struggle to articulate them clearly. However, as the GPT builder assists in eliciting their inherent needs through questioning, designers can explore users' unexpressed perceptions about conversational AI.

3.2.2 Specific Requirements Emerged While Using the Personalized ChatGPT. There have been many attempts to explore users' intrinsic thoughts through generative sessions before. However, in most cases, developing and assessing AI services was a separate process. However, participants can immediately use the newly created personalized ChatGPT in the preview section of the GPT-creating tool. When conversing with the completed personalized ChatGPT, participants request specific requirements not asked for during the building phase. It is possible to understand why users created such an agent or what their expectations are. To give an example, P3, who created a traveling assistant, asked ChatGPT "Check my wearables for my sleep and other statuses" and "Find the reservation number in my email and cancel the hotel booking." The user's requirements have been concretized into features providing direct assistance, such as accessing health data or personal information and rescheduling tasks based on the situation.

3.2.3 Able to Discover Their Subconscious Desire Through Modification. The GPT-creating tool suggests modification if needed after experiencing the personalized ChatGPT. Participants undergo an iterative process of using the personalized ChatGPT, making modifications, and refining it according to their preferences. To illustrate, P11 attempted to converse with the personalized ChatGPT by acting as if she were her mother, who exhibits symptoms of dementia. When ChatGPT focused on providing solutions rather than empathetic dialogue, P11 requested the GPT builder to "focus more on empathizing rather than just offering solutions." Similarly, when the cooking assistant created by P4, asked the user for information all at once (such as cooking skills, available ingredients, budget, and preferred styles), he requested the GPT builder to "ask questions one by one, in a step by step manner."

Remarkably, sometimes, participants' expectations regarding the role of the agent changed as they interacted with personalized ChatGPT. P7 expected a task-oriented role, hoping that ChatGPT would refine her expression to be more sophisticated and logical in sharing her ideas with others for the first time. However, following interactions with the personalized ChatGPT, P7 adjusted her expectations from a task-oriented role to a social one, expressing a desire to engage in in-depth, social conversations with it. Sometimes, users may not even know what they desire, referred to as latent needs [11], and they are often challenging to identify. However, as users engage in the iterative process with the GPT-creating tool, they begin to specify preferences, thereby revealing inherent needs of which they were previously unaware.

4 LIMITATIONS AND FUTURE WORKS

In this study, we explore opportunities of utilizing a GPT-creating tool to uncover intrinsic user needs and expectations through generative sessions. Our research discovered the opportunity of the GPT-creating tool to facilitate the specification of users' latent subconscious ideas previously abstracted in their minds.

However, it is important to note the limitation that we did not develop an innovative tool for exploring intrinsic needs but rather utilized the existing GPT-creating tool released by OpenAI. Since the GPT-creating tool is also based on the LLM system, it sometimes behaved unexpectedly. For instance, there was an occasion when the GPT builder did not ask a participant for specific instructions and directly finalized creating a personalized ChatGPT. To overcome these limitations, further research can be conducted on design tools for exploring users' needs.

Furthermore, our study specifically targeted middle-aged adults to discover the value of the using GPT-creating tool as a generative session. Yet, the advantage of this tool is that it allows users with limited knowledge of AI to create their desired personalized ChatGPT through simple dialogues, as long as they can use a computer. Therefore, we anticipate that the GPT-creating tool could be effectively utilized in future research to explore inherent AI-related needs among various users with low AI literacy or who find it challenging to express latent thoughts, such as the elderly, young children, or non-verbal individuals with disabilities.

Lastly, as our goal was to discover the opportunities of the GPT-creating tool and considering that the middle-aged group was unfamiliar with using it, we experimented with a one-on-one setting to closely examine participants' usage of this tool. However, for subsequent research aimed at exploring the inherent needs of specific user groups, it would be feasible to conduct group participatory design sessions to explore a broader range of intrinsic needs.

5 CONCLUSION

In this study, we explored the usage of the GPT-creating tool as a generative approach through participatory design workshops with middle-aged individuals. We identified a range of inherent needs and expectations for AI support among middle-aged participants, which were not explicitly verbalized. Our analysis highlighted the strengths of using the GPT-creating tool as a design approach, particularly in uncovering unexpressed needs and expectations of users. The interactive nature of the GPT builder facilitated the concretization of unexpressed needs. Furthermore, the iterative use and modification of the GPT models helped in revealing users' subconscious needs. Since the GPT-creating tool has proven to be a viable approach for generative sessions that support designers in understanding users' intrinsic needs and expectations, we anticipate its widespread use in future research to explore the inherent needs of various user groups regarding AI.

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